Lecture 4 (9/21)

Continuous Distributions

The uniform distribution

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- but base on this formula $P(Y = y) = 0$ for all $y \in [a, b]$!
- Therefore, we cannot define a probability distribution by assigning outcomes in the usual way!

Question

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- What is the probability I wait more than 2 minutes?

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- What is the probability I wait more than 2 minutes?
- What is the probability I wait between 4 and 5 minutes, given that I waited more than 1 minute?

Densities

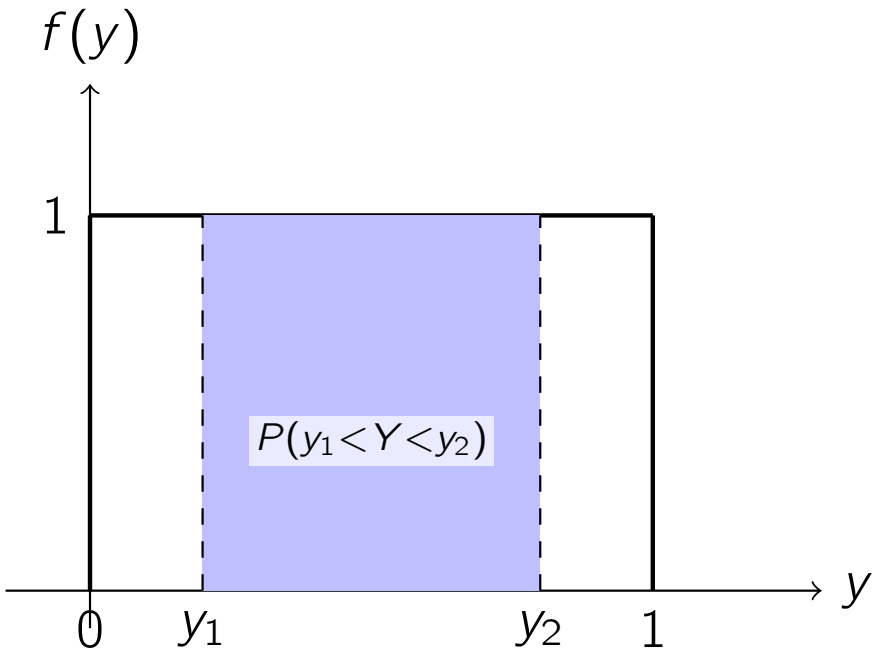
- We can think of the probabilities for a Uniform distribution as areas under a horizontal line of height $1/(b-a)$ between a and b.

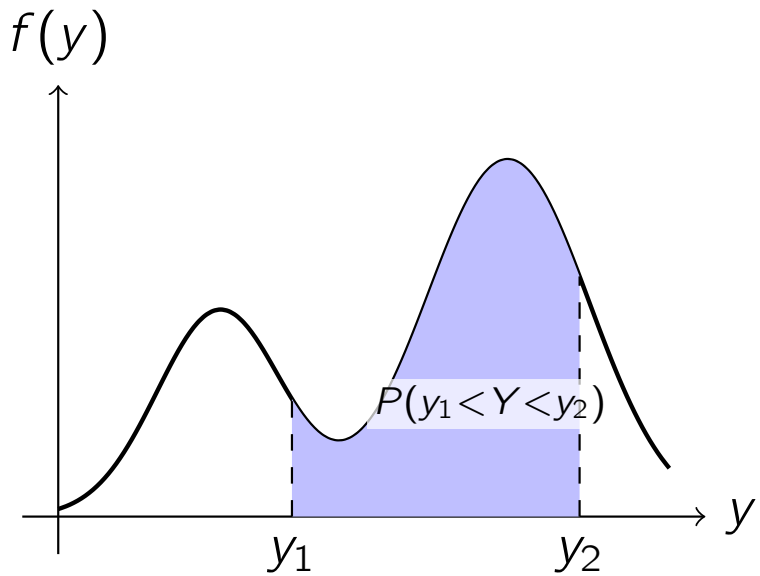
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Densities

A continuous random variable Y has a density $f(y) \geq 0$ with

$$P(a < Y < b) = \int_a^b f(y)\,dy, \qquad \int_{-\infty}^{\infty} f(y)\,dy = 1.$$

Note $f(y)$ can exceed 1; the only requirement is that the total area under f equals 1.

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- The most important thing for this class is to understand conceptually how probability calculations work, rarely do we evaluate anything but the simplest integral.

Density vs. CDF

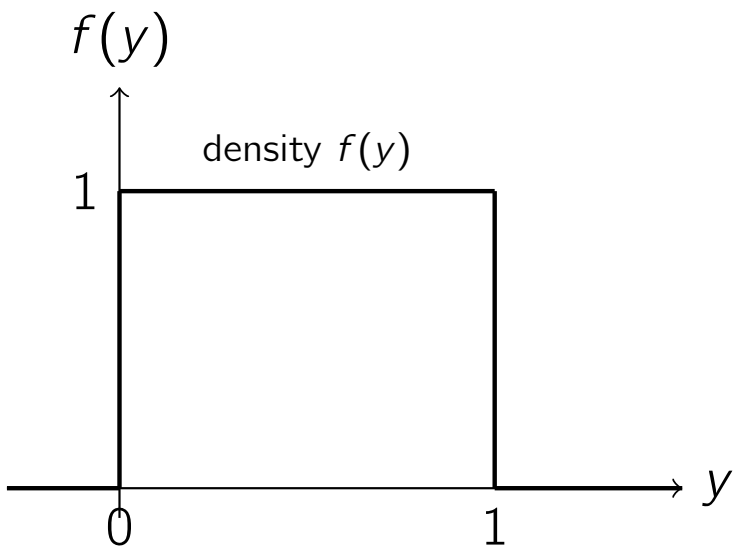
- The density $f(y)$ tells us how probability is spread out: probabilities come from *areas* under f.
- The CDF (cumulative distribution function) accumulates that area from the left:

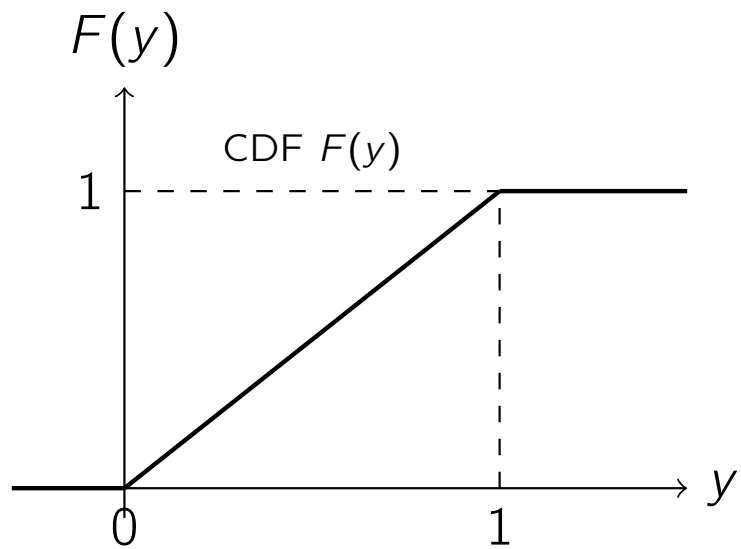
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Question

We often use $\text{Uniform}(0, 1)$ to model an unknown probability. Suppose a soccer team is about to take a penalty kick, and we want to bet on whether they score. We don't know the team's true scoring percentage q, so we model this as

$$q \sim \text{Uniform}(0, 1)$$
$$Y|q \sim \text{Bernoulli}(q)$$

What is the marginal distribution of Y? What is the density of $q|Y$?

- To find the marginal and conditional note that

$$P(Y = 1|q) = q$$

That is, if we know what q is, the Y is Bernoulli with probability of success q.[1]

- The density of q is $f(q) = 1$ when $q \in [0, 1]$ and marginalization works in the same way but with sums replaced with integrals:

$$P(Y = 1) = \int_0^1 P(Y = 1|q)f(q)dq = \int_0^1 qdq = \frac{1}{2}q^2\Big|_{q=0}^{q=1} = \frac{1}{2}$$

Therefore $Y \sim \text{Bernoulli}(1/2)$

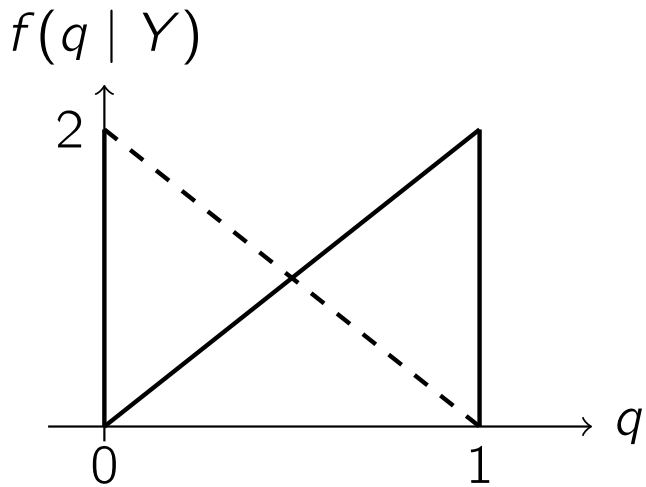
- and using Bayes' (or just the definition of conditional probability)

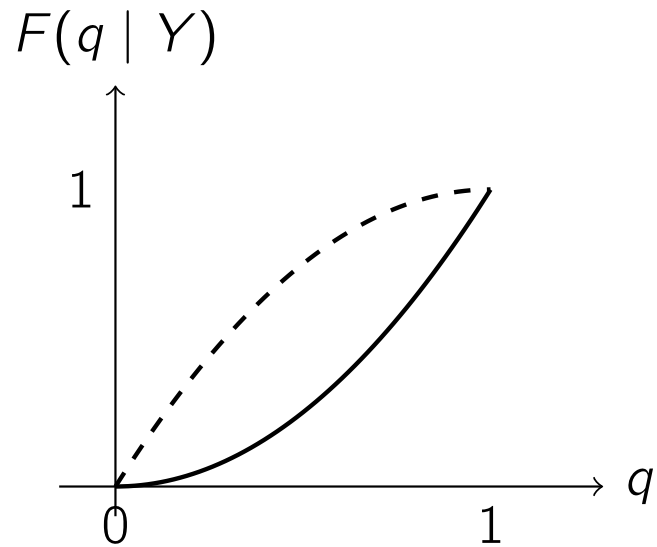
$$f(q|Y = 1) = \frac{P(Y = 1|q)f(q)}{P(Y = 1)} = 2q$$

[1] Technically we are conditioning on a set of measure zero, which has problems, but we will sweep this under the rug

$q|Y$

$$f(q \mid Y{=}1) = 2q, \qquad f(q \mid Y{=}0) = 2(1-q), \qquad q \in (0,1).$$





solid: $Y{=}1$ dashed: $Y{=}0$

Question

Let $U \sim \text{Uniform}(0, 1)$ and find the CDF and densities of U^2 and $\sqrt{U}$.

Question

Let $U \sim \text{Uniform}(0, 1)$ and define $T = -\frac{1}{\lambda} \log(U)$. Find the CDF and density of T.

Solutions

Bus stop, $T \sim \text{Uniform}(0, 5)$:

- $P(T > 2) = 3/5 = 0.6$
- $E[T \mid T > 3] = 4$ (since $T \mid T{>}3 \sim \text{Uniform}(3, 5)$)

$X \sim \text{Bernoulli}(1/4)$, $Y \mid X \sim \text{Uniform}(0, 1 + X)$:

- $E[Y] = 5/8 = 0.625$
- $P(Y = 1 \mid X > 1/2) = 0$ (continuous, so $P(Y{=}y) = 0$ for any y)

Soccer penalty kick, $q \sim \text{Uniform}(0, 1)$, $Y \mid q \sim \text{Bernoulli}(q)$:

- Marginal: $Y \sim \text{Bernoulli}(1/2)$
- $f(q \mid Y{=}1) = 2q$
- $f(q \mid Y{=}0) = 2(1 - q)$

$U \sim \text{Uniform}(0, 1)$:

- U^2: CDF $= \sqrt{v}$, density $= \dfrac{1}{2\sqrt{v}}$ on $(0, 1)$
- $\sqrt{U}$: CDF $= w^2$, density $= 2w$ on $(0, 1)$
- $T = -\log(U)/\lambda$: CDF $= 1 - e^{-\lambda t}$, density $= \lambda e^{-\lambda t}$, i.e. $T \sim \text{Exponential}(\lambda)$